

Phase 8 Aggregate Report

Overview

- Run count: 20.
- Condition count: 8.
- Best held-out TSPLIB condition: phase8_best_single_fixed_heuristic.

Condition Means

Condition	Final TSPLIB Gap	Final Transfer Gap	Selector Regret	Runtime-Adjusted Gap	Runtime Inflation	Mean Novelty	Mean Complexity
phase8_best_single_fixed_heuristic	0.07226	0.031222	0.0	0.07226	0.0	0.0	0.0
phase8_full_solver_evolution	0.167249	0.11025	0.094989	0.167249	-0.79877	0.732834	0.66
phase8_llm_evolved_adaptive_controller	0.155397	0.083357	0.083137	0.27381	0.746118	0.283303	0.897125
phase8_llm_evolved_controller_diversity_failure_replay	0.16551	0.093285	0.09325	0.337175	1.054372	0.503036	0.846875
phase8_llm_static_selector	0.193515	0.095613	0.121255	0.295981	0.309192	0.0	0.746
phase8_oracle_selector	0.07226	0.026678	0.0	0.07338	0.011291	0.0	0.0
phase8_random_portfolio	0.191943	0.122316	0.119683	0.258686	-0.107764	0.0	0.0
phase8_supervised_ml_selector	0.07226	0.031222	0.0	0.074051	0.020029	0.0	0.0

Paired Comparisons

random_vs_best_fixed

- Candidate: phase8_random_portfolio versus reference phase8_best_single_fixed_heuristic across 20 paired runs.
- Held-out TSPLIB gap delta: 0.119683 (95% CI 0.108269 to 0.130787).
- Transfer gap delta: 0.091094 (95% CI 0.082606 to 0.099808).
- Selector regret delta: 0.119683 (95% CI 0.10817 to 0.130823).

supervised_vs_best_fixed

- Candidate: phase8_supervised_ml_selector versus reference phase8_best_single_fixed_heuristic across 20 paired runs.
- Held-out TSPLIB gap delta: 0.0 (95% CI 0.0 to 0.0).
- Transfer gap delta: 0.0 (95% CI 0.0 to 0.0).
- Selector regret delta: 0.0 (95% CI 0.0 to 0.0).

static_vs_supervised

- Candidate: phase8_llm_static_selector versus reference phase8_supervised_ml_selector across 20 paired runs.
- Held-out TSPLIB gap delta: 0.121255 (95% CI 0.099278 to 0.140226).
- Transfer gap delta: 0.064391 (95% CI 0.049635 to 0.078397).
- Selector regret delta: 0.121255 (95% CI 0.099839 to 0.140491).

adaptive_vs_static

- Candidate: phase8_llm_evolved_adaptive_controller versus reference phase8_llm_static_selector across 20 paired runs.
- Held-out TSPLIB gap delta: -0.038118 (95% CI -0.06868 to -0.00707).
- Transfer gap delta: -0.012256 (95% CI -0.033612 to 0.008575).
- Selector regret delta: -0.038118 (95% CI -0.068547 to -0.006883).

replay_vs_adaptive

- Candidate: phase8_llm_evolved_controller_diversity_failure_replay versus reference phase8_llm_evolved_adaptive_controller across 20 paired runs.
- Held-out TSPLIB gap delta: 0.010114 (95% CI -0.020641 to 0.041674).
- Transfer gap delta: 0.009928 (95% CI -0.010223 to 0.031181).
- Selector regret delta: 0.010114 (95% CI -0.02087 to 0.041393).

full_solver_vs_adaptive

- Candidate: phase8_full_solver_evolution versus reference phase8_llm_evolved_adaptive_controller across 20 paired runs.
- Held-out TSPLIB gap delta: 0.011853 (95% CI -0.025032 to 0.049041).
- Transfer gap delta: 0.026893 (95% CI -0.004858 to 0.059364).
- Selector regret delta: 0.011853 (95% CI -0.024976 to 0.04907).

oracle_vs_best_fixed

- Candidate: phase8_oracle_selector versus reference phase8_best_single_fixed_heuristic across 20 paired runs.
- Held-out TSPLIB gap delta: 0.0 (95% CI 0.0 to 0.0).
- Transfer gap delta: -0.004544 (95% CI -0.004544 to -0.004544).
- Selector regret delta: 0.0 (95% CI 0.0 to 0.0).

Portfolio Answers

- Selector progress: Supervised selection changed held-out TSPLIB gap by 0.0 versus the best fixed heuristic. Adaptive LLM control changed held-out TSPLIB gap by -0.038118 versus the static LLM selector, while the oracle remained 0.0 better than the best fixed heuristic.
- Replay effect: Diversity-failure replay changed held-out TSPLIB gap by 0.010114 and selector regret by 0.010114 relative to the no-replay adaptive controller.
- Interpretability tradeoff: Full-solver evolution changed held-out TSPLIB gap by 0.011853 and runtime-adjusted gap by -0.106561 relative to the adaptive controller.